Class 2

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Today we have two datasets to explore. We'll continue our exploration using R to understand and visualize data.

Part 1: Donuts

Make sure you have opened the Rproject and that the rmd and data files are in the same folder.

Load the Simpson's dataset from the book chapter assigned this week.

```
load("donutd.RData")
```

Looking over the data

```
head(donutd)
```

```
##          name donuts weight child male donutsXnew
## 1       Homer  14.00    275     0    0      14.00
## 2       Marge   0.00    141     0    1       0.00
## 3        Lisa   0.00     70    -1    1       0.00
## 4        Bart   5.00     75    -1    0       5.00
## 5 ComicBookGuy  20.00    310     0    0      20.00
## 6    Mr Burns   0.75     80     0    0       0.75
```

```
tail(donutd)
```

```
##                 name donuts weight child male donutsXnew
## 8       Chief Wiggum   16.0    263     0    0       16.0
## 9  Principal Skinner    3.0    205     0    0        3.0
## 10      Rev. Lovejoy    2.0    185     0    0        2.0
## 11      Ned Flanders    0.8    170     0    0        0.8
## 12             Patty    5.0    155     0    1        5.0
## 13             Selma    4.0    145     0    1        4.0
```

```
summary(donutd)
```

```
##         name           donuts           weight          child        
##  Length    :13   Min.   : 0.000   Min.   : 70.0   Min.   :-1.0000  
##  N.unique :13   1st Qu.: 0.750   1st Qu.:141.0   1st Qu.: 0.0000  
##  N.blank   : 0   Median : 3.000   Median :160.0   Median : 0.0000  
##  Min.nchar: 4   Mean   : 5.446   Mean   :171.8   Mean   :-0.1538  
##  Max.nchar:17   3rd Qu.: 5.000   3rd Qu.:205.0   3rd Qu.: 0.0000  
##                 Max.   :20.000   Max.   :310.0   Max.   : 0.0000  
##       male          donutsXnew    
##  Min.   :0.0000   Min.   : 0.000  
##  1st Qu.:0.0000   1st Qu.: 0.750  
##  Median :0.0000   Median : 3.000  
```

```
##  Mean   :0.3077   Mean   : 5.446
##  3rd Qu.:1.0000   3rd Qu.: 5.000
##  Max.   :1.0000   Max.   :20.000
```

```r
dim(donutd) #Count of how many observations and variables
```

```
## [1] 13  6
```

Male and name are not continuous(numeric in R speak) variables.They are categorical and we can get counts in each category.

frequency tables:

```r
table(donutd$male) #gives count of different categories
```

```
##
## 0 1
## 9 4
```

```r
table(donutd$name)
```

```
##
##              Bart      Chief Wiggum      ComicBookGuy             Homer
##                 1                 1                 1                 1
##              Lisa             Marge          Mr Burns      Ned Flanders
##                 1                 1                 1                 1
##             Patty Principal Skinner      Rev. Lovejoy             Selma
##                 1                 1                 1                 1
##         Smithers
##                 1
```

```r
prop.table(table(donutd$male)) #gives percentages
```

```
##
##         0         1
## 0.6923077 0.3076923
```

I use this command all the time to get a sense of the data.

What percentage of the individuals in the dataset are children?

```r
prop.table(table(donutd$child))
```

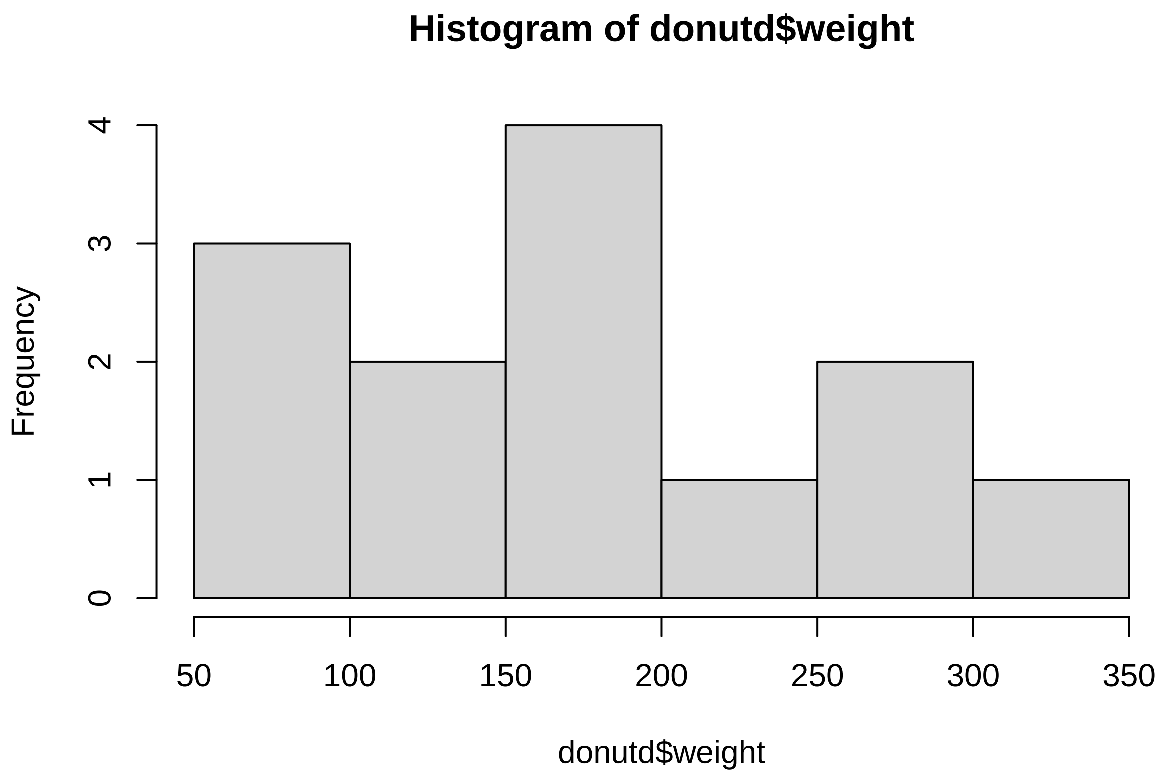
```
##
##        -1         0
## 0.1538462 0.8461538
```

More review:

```r
summary(donutd$weight)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    70.0   141.0   160.0   171.8   205.0   310.0
```

```r
hist(donutd$weight)
```

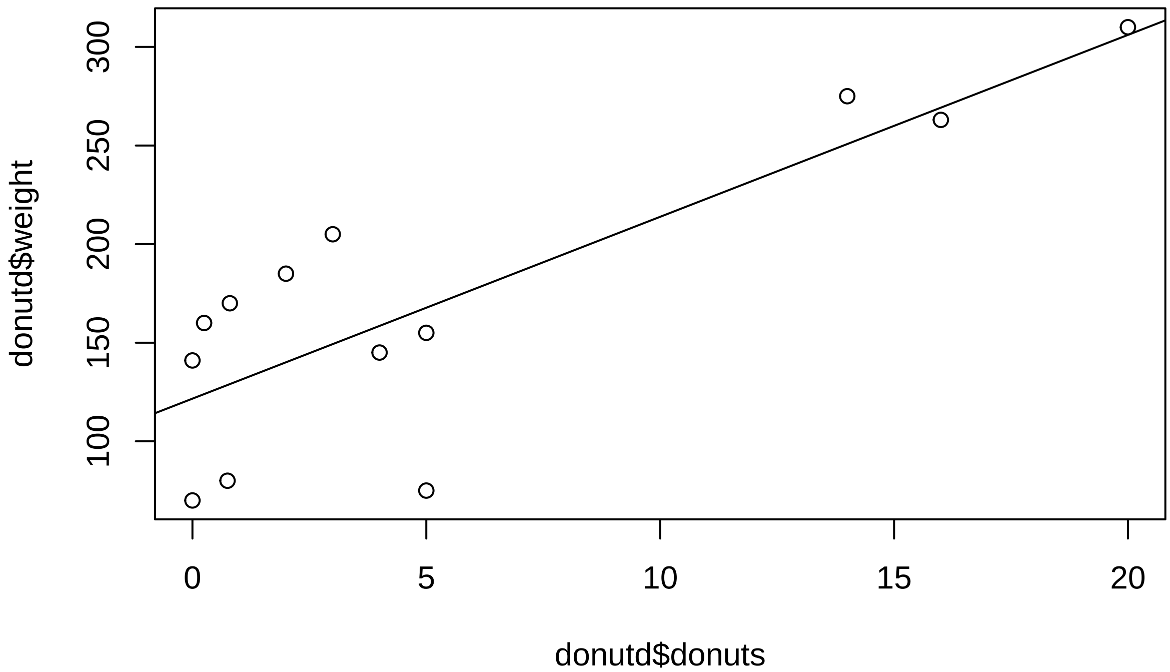



```r
# check-in:  What is the average number of donuts consumed? Who eats the most?
summary(donutd$donuts)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   0.750   3.000   5.446   5.000  20.000
```

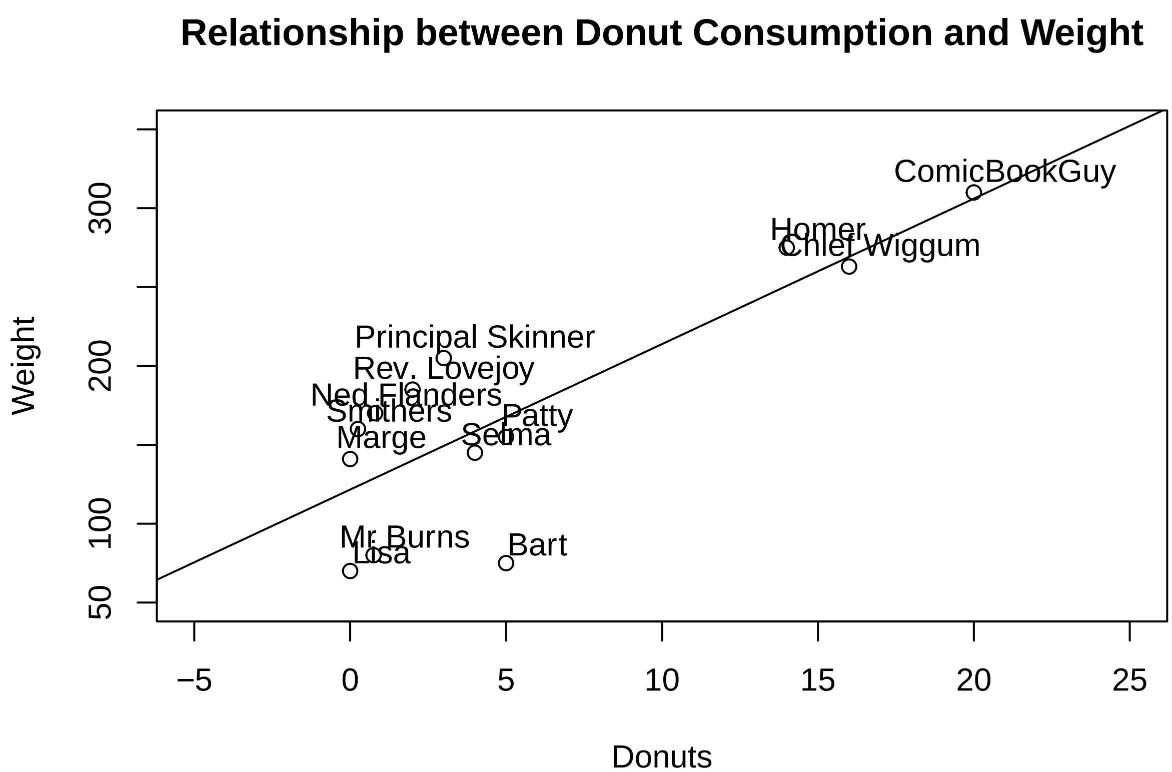
Plot the relationship between donuts and weight

```r
plot(donutd$donuts, donutd$weight) #scatterplot (X then Y)
abline(lm(donutd$weight~donutd$donuts)) #add trend line (Y then X)
```



Make plot look a bit more professional

```
plot(donutd$donuts, donutd$weight, main="Relationship between Donut Consumption and Weight", xlab="Donu
abline(lm(donutd$weight~donutd$donuts)) #and boom, you've already ran your 2nd regression
text(donutd$donuts+1, donutd$weight+12, donutd$name)#add labels(the -12 is to offset the label)
```

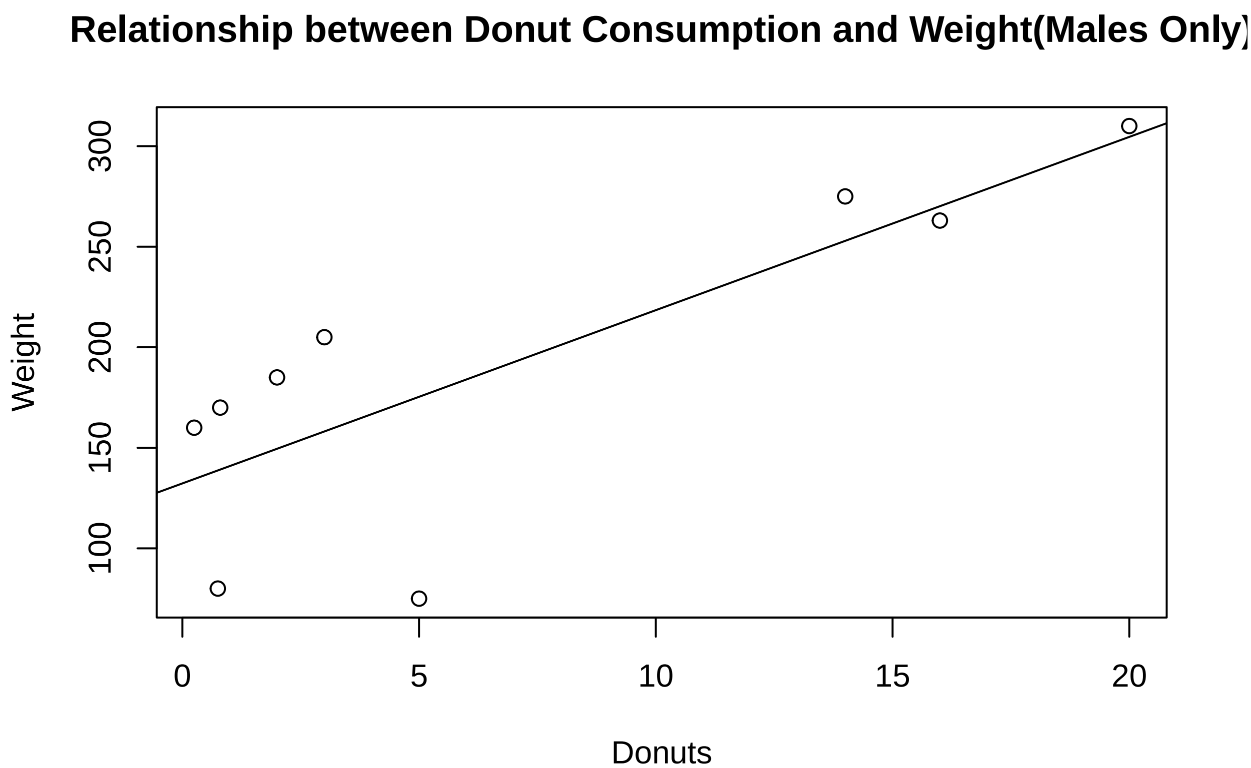


Does there appear to be a relationship in the sample?

What about with just males? There are multiple ways to do this:

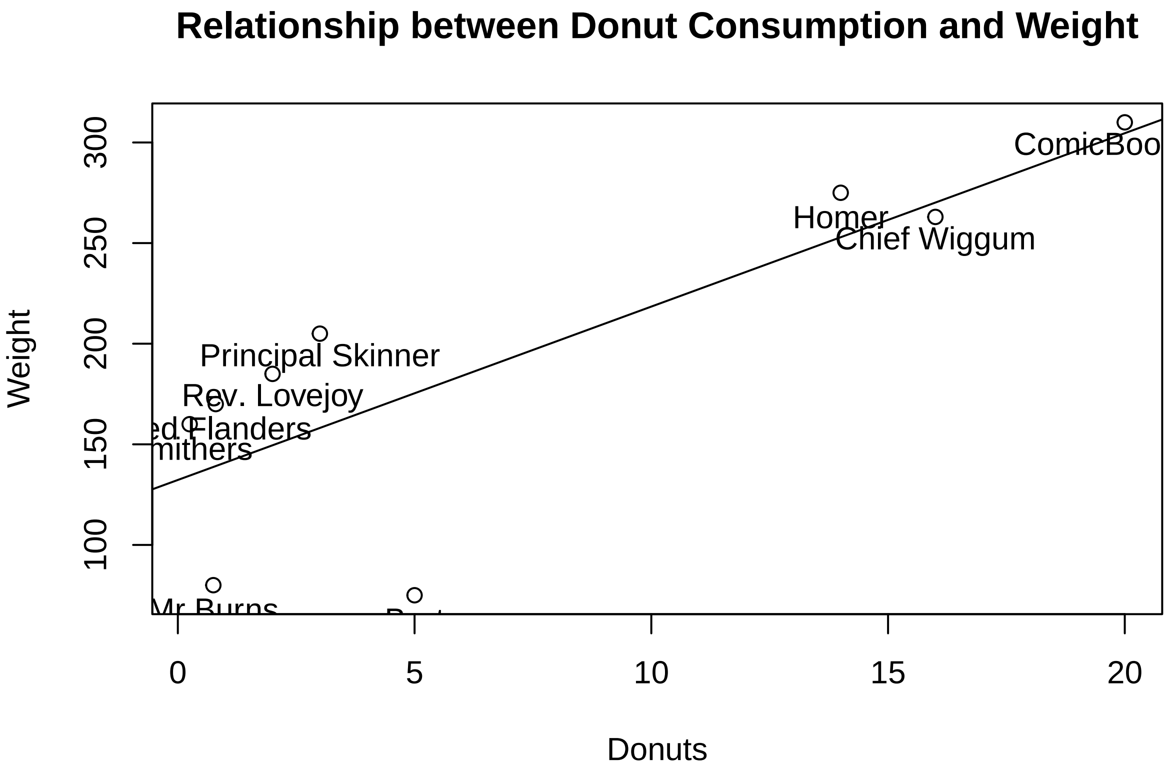
First off, if we don't think we'll reuse the dataset we can pull out what using []

```
plot(donutd$donuts[donutd$male==0], donutd$weight[donutd$male==0], main="Relationship between Donut Cons
abline(lm(donutd$weight[donutd$male==0]~donutd$donuts[donutd$male==0])) #
```



Or we can create a new dataset ie a subset(I almost choose this option)

```
males <- donutd[which(donutd$male==0),]

plot(males$donuts, males$weight, main="Relationship between Donut Consumption and Weight", xlab="Donuts
abline(lm(males$weight~males$donuts)) #and boom, you've already ran your 2nd regression
text(males$donuts, males$weight-12, males$name)
```



Heads Up: We will practice this a third way using the dplyr package in the coming weeks

Exclude kids

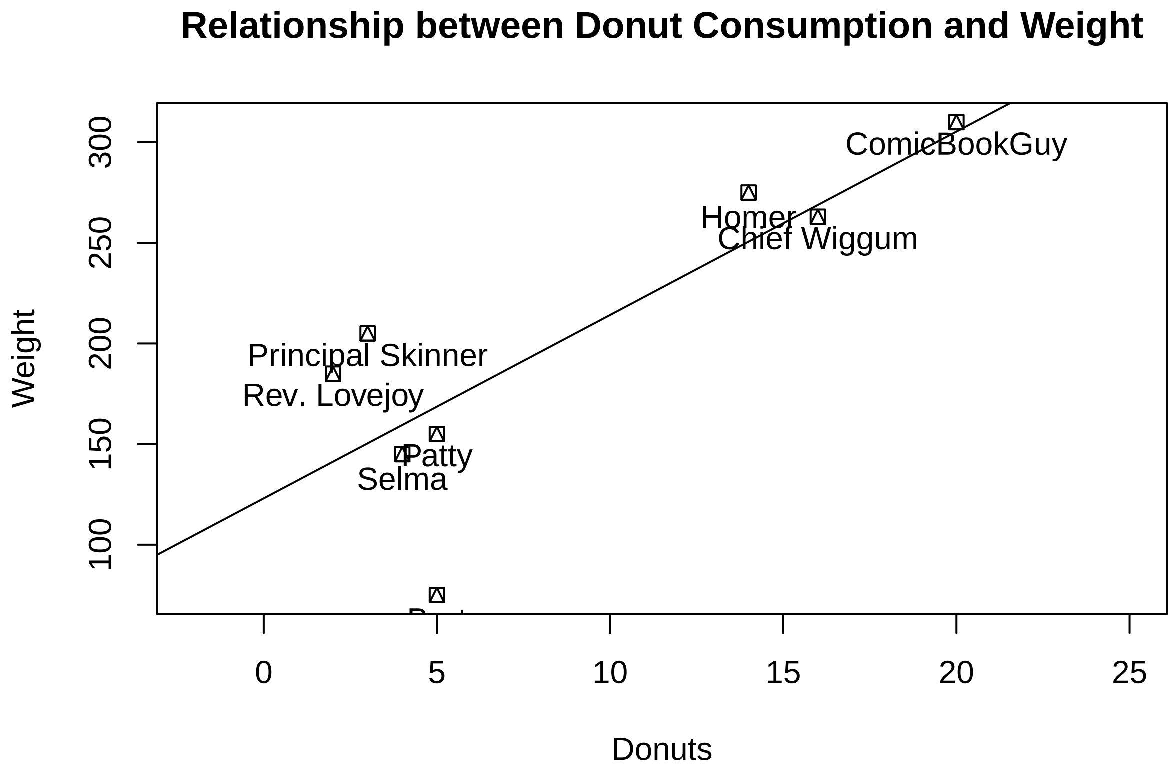
```
adults <- donutd[which(donutd$child!=-1),] # the != does not equal
```

Exclude overweight - use greater than or less than

```
normweight <- donutd[which(donutd$weight< 250),] # less than 250
```

Try creating a scatterplot with just the 1) females OR 2) characters who at least eat donuts.

```
eaters <- donutd[which(donutd$donuts > 1),]

plot(eaters$donuts, eaters$weight, main="Relationship between Donut Consumption and Weight", xlab="Donu
abline(lm(eaters$weight~eaters$donuts)) #and boom, you've already ran your 2nd regression
text(eaters$donuts, eaters$weight-12, eaters$name)
```



Part 3: Congressional Age

#Looking at a different dataset: Time to look at something a bit more fun. The age of congress. The data are from this article: https://fivethirtyeight.com/features/aging-congress-boomers/ from a few years ago.

We want to load a csv file into R. Here's how you do it. Make that sure when you do this, your data is in the same working directory that your project is in. It is easy save excel files as csv.

```
oldc <- read.csv("data_aging_congress.csv", header=T)
```

Different file formats have different load commands. If your data is already saved as an .RData file, then run this code:load("name_of_data_set.RData")

We're not going to run it here, because we only have a CSV dataset.R can load excel, spss, stata, text, etc. if you use the right command. For any data you find online you should be able to load it into R after some quick googling.

In R studio it is easy to load data by clicking it on it using the files window(bottom right). This is great, however you need to put it in your code or it will not knit

As usual, ook at the first and last 5 lines of the data frame

```
head(oldc)
```

```
##   congress start_date chamber state_abbrev party_code                 bioname
## 1       82 1951-01-03   House           ND        200    AANDAHL, Fred George
## 2       80 1947-01-03   House           VA        100 ABBITT, Watkins Moorman
## 3       81 1949-01-03   House           VA        100 ABBITT, Watkins Moorman
## 4       82 1951-01-03   House           VA        100 ABBITT, Watkins Moorman
## 5       83 1953-01-03   House           VA        100 ABBITT, Watkins Moorman
## 6       84 1955-01-03   House           VA        100 ABBITT, Watkins Moorman
##   bioguide_id   birthday cmltv_cong cmltv_chamber age_days age_years generation
```

```
## 1     A000001 1897-04-09         1            1    19626  53.73306       Lost
## 2     A000002 1908-05-21         1            1    14106  38.62012   Greatest
## 3     A000002 1908-05-21         2            2    14837  40.62149   Greatest
## 4     A000002 1908-05-21         3            3    15567  42.62012   Greatest
## 5     A000002 1908-05-21         4            4    16298  44.62149   Greatest
## 6     A000002 1908-05-21         5            5    17028  46.62012   Greatest
```

```r
tail(oldc)
```

```
##       congress start_date chamber state_abbrev party_code       bioname
## 29115      115 2017-01-03   House           NY        200 ZELDIN, Lee M
## 29116      116 2019-01-03   House           NY        200 ZELDIN, Lee M
## 29117      117 2021-01-03   House           NY        200 ZELDIN, Lee M
## 29118      114 2015-01-03   House           MT        200   ZINKE, Ryan
## 29119      115 2017-01-03   House           MT        200   ZINKE, Ryan
## 29120      118 2023-01-03   House           MT        200   ZINKE, Ryan
##       bioguide_id   birthday cmltv_cong cmltv_chamber age_days age_years
## 29115     Z000017 1980-01-30          2             2    13488  36.92813
## 29116     Z000017 1980-01-30          3             3    14218  38.92676
## 29117     Z000017 1980-01-30          4             4    14949  40.92813
## 29118     Z000018 1961-11-01          1             1    19421  53.17180
## 29119     Z000018 1961-11-01          2             2    20152  55.17317
## 29120     Z000018 1961-11-01          3             3    22343  61.17180
##       generation
## 29115      Gen X
## 29116      Gen X
## 29117      Gen X
## 29118    Boomers
## 29119    Boomers
## 29120    Boomers
```

```r
str(oldc) #This tells us what kind of data R thinks each variable is - we'll talk about this later
```

```
## 'data.frame':    29120 obs. of  13 variables:
##  $ congress     : int  82 80 81 82 83 84 85 86 87 88 ...
##  $ start_date   : chr  "1951-01-03" "1947-01-03" "1949-01-03" "1951-01-03" ...
##  $ chamber      : chr  "House" "House" "House" "House" ...
##  $ state_abbrev : chr  "ND" "VA" "VA" "VA" ...
##  $ party_code   : int  200 100 100 100 100 100 100 100 100 100 ...
##  $ bioname      : chr  "AANDAHL, Fred George" "ABBITT, Watkins Moorman" "ABBITT, Watkins Moorman" "AI
##  $ bioguide_id  : chr  "A000001" "A000002" "A000002" "A000002" ...
##  $ birthday     : chr  "1897-04-09" "1908-05-21" "1908-05-21" "1908-05-21" ...
##  $ cmltv_cong   : int  1 1 2 3 4 5 6 7 8 9 ...
##  $ cmltv_chamber: int  1 1 2 3 4 5 6 7 8 9 ...
##  $ age_days     : int  19626 14106 14837 15567 16298 17028 17759 18489 19220 19950 ...
##  $ age_years    : num  53.7 38.6 40.6 42.6 44.6 ...
##  $ generation   : chr  "Lost" "Greatest" "Greatest" "Greatest" ...
```

Look at variable names

```r
names(oldc)
```

```
##  [1] "congress"      "start_date"    "chamber"       "state_abbrev"
##  [5] "party_code"    "bioname"       "bioguide_id"   "birthday"
##  [9] "cmltv_cong"    "cmltv_chamber" "age_days"      "age_years"
## [13] "generation"
```

Number of rows in your data? What is our unit of analysis?

```r
nrow(oldc)
```

```
## [1] 29120
```

Dimensions of your data? # Observations then # of Variables

```r
dim(oldc)
```

```
## [1] 29120    13
```

congress and generation

```r
table(oldc$congress)
```

```
## 
##  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80  81  82  83  84  85 
## 555 560 557 548 551 567 563 547 549 554 561 561 554 564 553 553 553 552 540 545 
##  86  87  88  89  90  91  92  93  94  95  96  97  98  99 100 101 102 103 104 105 
## 552 558 551 548 543 551 548 548 551 551 544 545 544 541 544 545 548 546 543 544 
## 106 107 108 109 110 111 112 113 114 115 116 117 118 
## 539 547 539 540 549 553 546 548 541 555 544 551 536 
```

```r
table(oldc$generation)
```

```
## 
##     Boomers       Gen X       Gen Z      Gilded    Greatest        Lost 
##        5108        1130           1          15        7147        4732 
##  Millennial  Missionary Progressive      Silent 
##         133        4768         485        5601 
```

```r
table(oldc$generation, oldc$congress)
```

```
## 
##               66  67  68  69  70  71  72  73  74  75  76  77  78  79  80  81
##   Boomers       0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Gen X         0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Gen Z         0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Gilded        5   4   3   1   1   1   0   0   0   0   0   0   0   0   0   0
##   Greatest      0   0   0   0   0   1   2   9  21  32  54  74  92 120 167 198
##   Lost         19  31  49  69  89 107 146 198 239 271 293 311 306 316 297 289
##   Millennial    0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Missionary  412 418 431 419 414 424 396 329 284 249 211 173 155 127  89  66
##   Progressive 119 107  74  59  47  34  19  11   5   2   3   3   1   1   0   0
##   Silent        0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##              
##               82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97
##   Boomers       0   0   0   0   0   0   0   0   0   0   0   0   4  10  18  40
##   Gen X         0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Gen Z         0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Gilded        0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Greatest    220 251 278 305 368 390 415 437 436 441 422 398 343 298 247 199
##   Lost        283 263 235 215 169 154 115  79  60  45  34  21  15   7   2   1
##   Millennial    0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Missionary   50  38  27  25  14   9   4   2   2   0   0   0   0   0   0   0
##   Progressive   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Silent        0   0   0   0   1   5  17  30  45  65  92 129 189 236 277 305
##              
```

```
##                98  99 100 101 102 103 104 105 106 107 108 109 110 111 112 113
##   Boomers      59  67  83  95 113 164 207 235 251 274 289 305 328 330 336 346
##   Gen X         0   0   0   0   0   0   3   6   7  12  19  25  35  55  80 101
##   Gen Z         0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Gilded        0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Greatest    176 155 130 111 104  68  48  35  29  22  14  11  10   7   6   3
##   Lost          1   1   1   1   0   0   0   0   0   0   0   0   0   0   0   0
##   Millennial    0   0   0   0   0   0   0   0   0   0   0   0   0   1   1   3
##   Missionary    0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Progressive   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
##   Silent      308 318 330 338 331 314 285 268 252 239 217 199 176 160 123  95
##
##               114 115 116 117 118
##   Boomers     340 351 303 301 259
##   Gen X       122 136 162 175 192
##   Gen Z         0   0   0   0   1
##   Gilded        0   0   0   0   0
##   Greatest      0   0   0   0   0
##   Lost          0   0   0   0   0
##   Millennial    4   6  26  37  55
##   Missionary    0   0   0   0   0
##   Progressive   0   0   0   0   0
##   Silent       75  62  53  38  29
```

```r
table(oldc$party_code)
```

```
##
##   100   112   200   328   329   331   347   356   370   380   402   522   523
## 15804     3 13181    44     3     2     1     1    29     4     1     7     1
##   537
##    39
```

Count unique representatives

```r
length(unique(oldc$bioguide_id))
```

```
## [1] 5257
```

```r
length(unique(oldc$bioname))
```

```
## [1] 5251
```

Summary of the data will give you mean, minimum, maximum values, etc.

```r
summary(oldc$age_years)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   23.67   45.81   53.45   53.73   61.22   98.08
```

```r
summary(oldc) # Runs for all numeric variables in the dataset
```

```
##     congress          start_date          chamber          state_abbrev
##  Min.   : 66.00   Length    :29120   Length    :29120   Length    :29120
##  1st Qu.: 79.00   N.unique  :   53   N.unique  :    2   N.unique  :   50
##  Median : 92.00   N.blank   :    0   N.blank   :    0   N.blank   :    0
##  Mean   : 91.88   Min.nchar:   10   Min.nchar:    5   Min.nchar:    2
##  3rd Qu.:105.00   Max.nchar:   10   Max.nchar:    6   Max.nchar:    2
##  Max.   :118.00
##    party_code         bioname          bioguide_id          birthday
```

```
##  Min.   :100.0   Length   :29120   Length   :29120   Length   :29120
##  1st Qu.:100.0   N.unique : 5251   N.unique : 5257   N.unique : 4935
##  Median :100.0   N.blank  :    0   N.blank  :    0   N.blank  :    0
##  Mean   :146.7   Min.nchar:    7   Min.nchar:    7   Min.nchar:   10
##  3rd Qu.:200.0   Max.nchar:   42   Max.nchar:    7   Max.nchar:   10
##  Max.   :537.0
##    cmltv_cong     cmltv_chamber      age_days       age_years
##  Min.   : 1.000   Min.   : 1.000   Min.   : 8644   Min.   :23.67
##  1st Qu.: 2.000   1st Qu.: 2.000   1st Qu.:16732   1st Qu.:45.81
##  Median : 4.000   Median : 4.000   Median :19523   Median :53.45
##  Mean   : 5.414   Mean   : 5.112   Mean   :19626   Mean   :53.73
##  3rd Qu.: 8.000   3rd Qu.: 7.000   3rd Qu.:22359   3rd Qu.:61.22
##  Max.   :30.000   Max.   :30.000   Max.   :35824   Max.   :98.08
##     generation
##  Length   :29120
##  N.unique :   10
##  N.blank  :    0
##  Min.nchar:    4
##  Max.nchar:   11
##
```

Mean age?

```r
mean(oldc$age_years)
```

```
## [1] 53.73248
```

Minimum of age?

```r
min(oldc$age_years)
```

```
## [1] 23.66598
```

Maximum age?

```r
max(oldc$age_years)
```

```
## [1] 98.08077
```

Range of ages in this data

```r
range(oldc$age_years)
```

```
## [1] 23.66598 98.08077
```

Check: What is the average cumulative time in congress(cmltv_cong)

Bonus check: Play around with the data and tell the class something interesting.

```r
table(oldc$congress, oldc$generation)
```

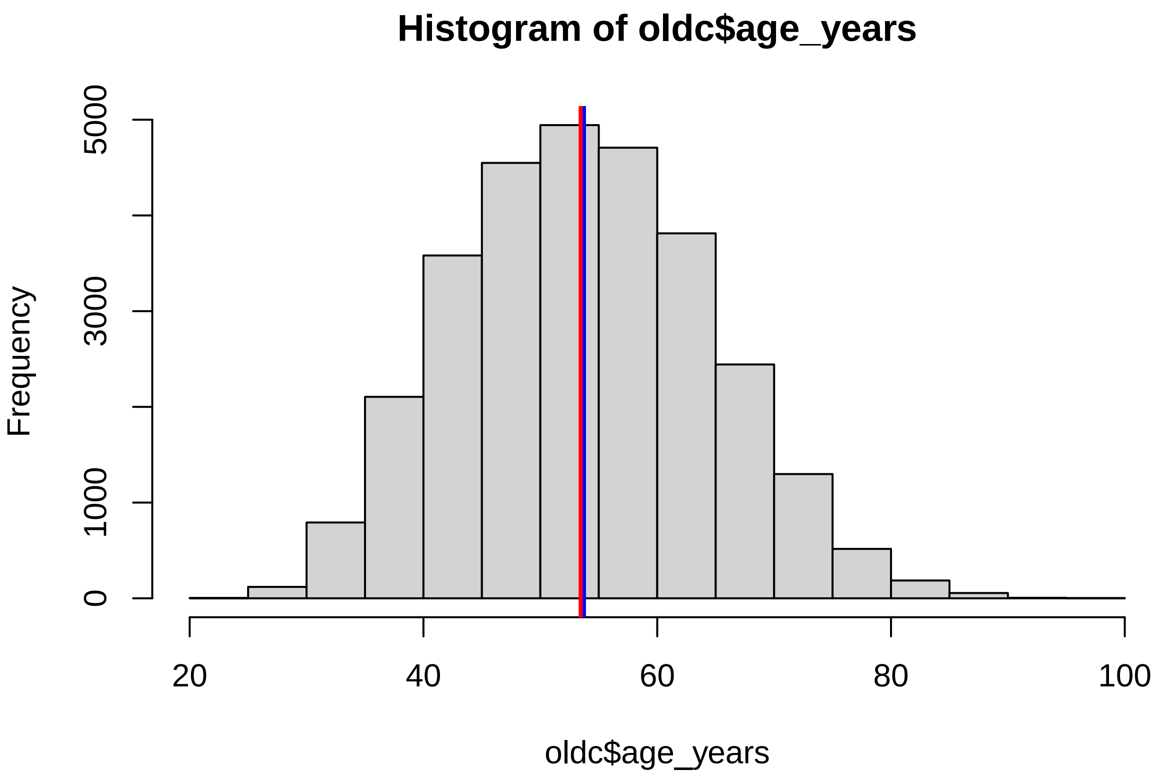
```
##
##      Boomers Gen X Gen Z Gilded Greatest Lost Millennial Missionary
##   66       0     0     0      5        0   19          0        412
##   67       0     0     0      4        0   31          0        418
##   68       0     0     0      3        0   49          0        431
##   69       0     0     0      1        0   69          0        419
##   70       0     0     0      1        0   89          0        414
##   71       0     0     0      1        1  107          0        424
##   72       0     0     0      0        2  146          0        396
##   73       0     0     0      0        9  198          0        329
```

```
##    74        0     0     0     0      21   239         0         284
##    75        0     0     0     0      32   271         0         249
##    76        0     0     0     0      54   293         0         211
##    77        0     0     0     0      74   311         0         173
##    78        0     0     0     0      92   306         0         155
##    79        0     0     0     0     120   316         0         127
##    80        0     0     0     0     167   297         0          89
##    81        0     0     0     0     198   289         0          66
##    82        0     0     0     0     220   283         0          50
##    83        0     0     0     0     251   263         0          38
##    84        0     0     0     0     278   235         0          27
##    85        0     0     0     0     305   215         0          25
##    86        0     0     0     0     368   169         0          14
##    87        0     0     0     0     390   154         0           9
##    88        0     0     0     0     415   115         0           4
##    89        0     0     0     0     437    79         0           2
##    90        0     0     0     0     436    60         0           2
##    91        0     0     0     0     441    45         0           0
##    92        0     0     0     0     422    34         0           0
##    93        0     0     0     0     398    21         0           0
##    94        4     0     0     0     343    15         0           0
##    95       10     0     0     0     298     7         0           0
##    96       18     0     0     0     247     2         0           0
##    97       40     0     0     0     199     1         0           0
##    98       59     0     0     0     176     1         0           0
##    99       67     0     0     0     155     1         0           0
##    100      83     0     0     0     130     1         0           0
##    101      95     0     0     0     111     1         0           0
##    102     113     0     0     0     104     0         0           0
##    103     164     0     0     0      68     0         0           0
##    104     207     3     0     0      48     0         0           0
##    105     235     6     0     0      35     0         0           0
##    106     251     7     0     0      29     0         0           0
##    107     274    12     0     0      22     0         0           0
##    108     289    19     0     0      14     0         0           0
##    109     305    25     0     0      11     0         0           0
##    110     328    35     0     0      10     0         0           0
##    111     330    55     0     0       7     0         1           0
##    112     336    80     0     0       6     0         1           0
##    113     346   101     0     0       3     0         3           0
##    114     340   122     0     0       0     0         4           0
##    115     351   136     0     0       0     0         6           0
##    116     303   162     0     0       0     0        26           0
##    117     301   175     0     0       0     0        37           0
##    118     259   192     1     0       0     0        55           0
##
##        Progressive Silent
##    66          119      0
##    67          107      0
##    68           74      0
##    69           59      0
##    70           47      0
##    71           34      0
##    72           19      0
```

```
##    73             11       0
##    74              5       0
##    75              2       0
##    76              3       0
##    77              3       0
##    78              1       0
##    79              1       0
##    80              0       0
##    81              0       0
##    82              0       0
##    83              0       0
##    84              0       0
##    85              0       0
##    86              0       1
##    87              0       5
##    88              0      17
##    89              0      30
##    90              0      45
##    91              0      65
##    92              0      92
##    93              0     129
##    94              0     189
##    95              0     236
##    96              0     277
##    97              0     305
##    98              0     308
##    99              0     318
##    100             0     330
##    101             0     338
##    102             0     331
##    103             0     314
##    104             0     285
##    105             0     268
##    106             0     252
##    107             0     239
##    108             0     217
##    109             0     199
##    110             0     176
##    111             0     160
##    112             0     123
##    113             0      95
##    114             0      75
##    115             0      62
##    116             0      53
##    117             0      38
##    118             0      29
```

Create a histogram for age in years

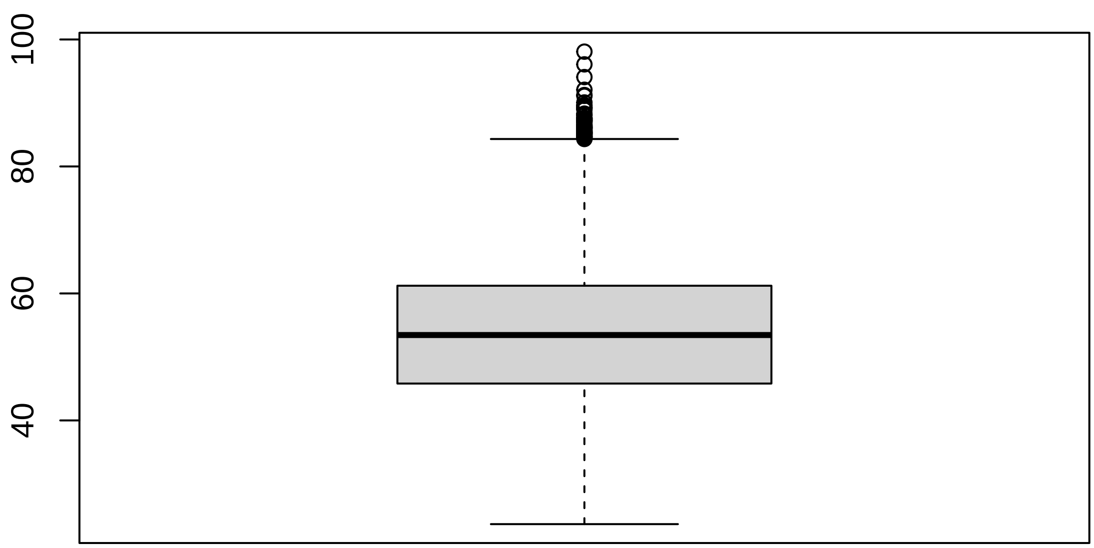
```
hist(oldc$age_years)
abline(v=mean(oldc$age_years), col="blue", lwd=2)
abline(v=median(oldc$age_years), col="red", lwd=2)
```



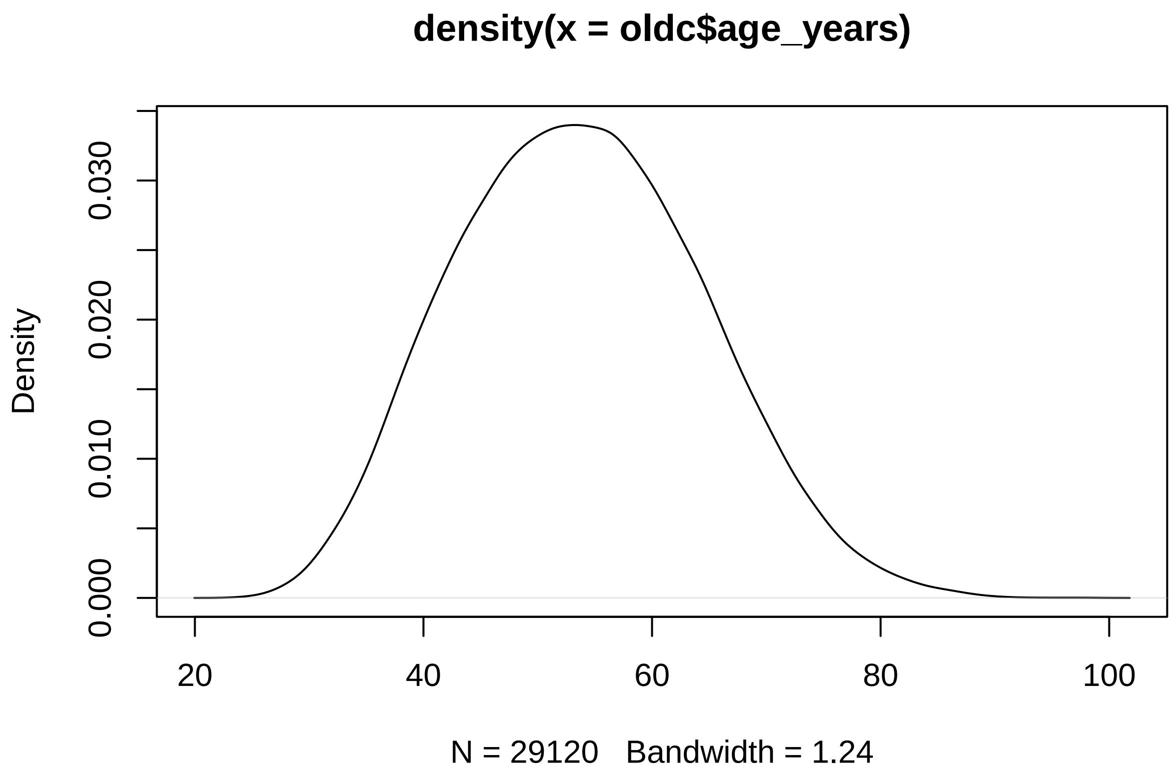
abline gives us a line. v is for vertical

Density plots and boxplots show us similar information

```
boxplot(oldc$age_years)
```




```
plot(density(oldc$age_years))
```




```
#How can we find out who the outliers are?
```

Check: How many democrats and republicans are in the sample

```
table(oldc$party_code)
```

```
##
##   100   112   200   328   329   331   347   356   370   380   402   522   523
## 15804     3 13181    44     3     2     1     1    29     4     1     7     1
##   537
##    39
```

```
mean(oldc$age_years[oldc$party_code==100]) #average for Democrats
```

```
## [1] 53.68089
```

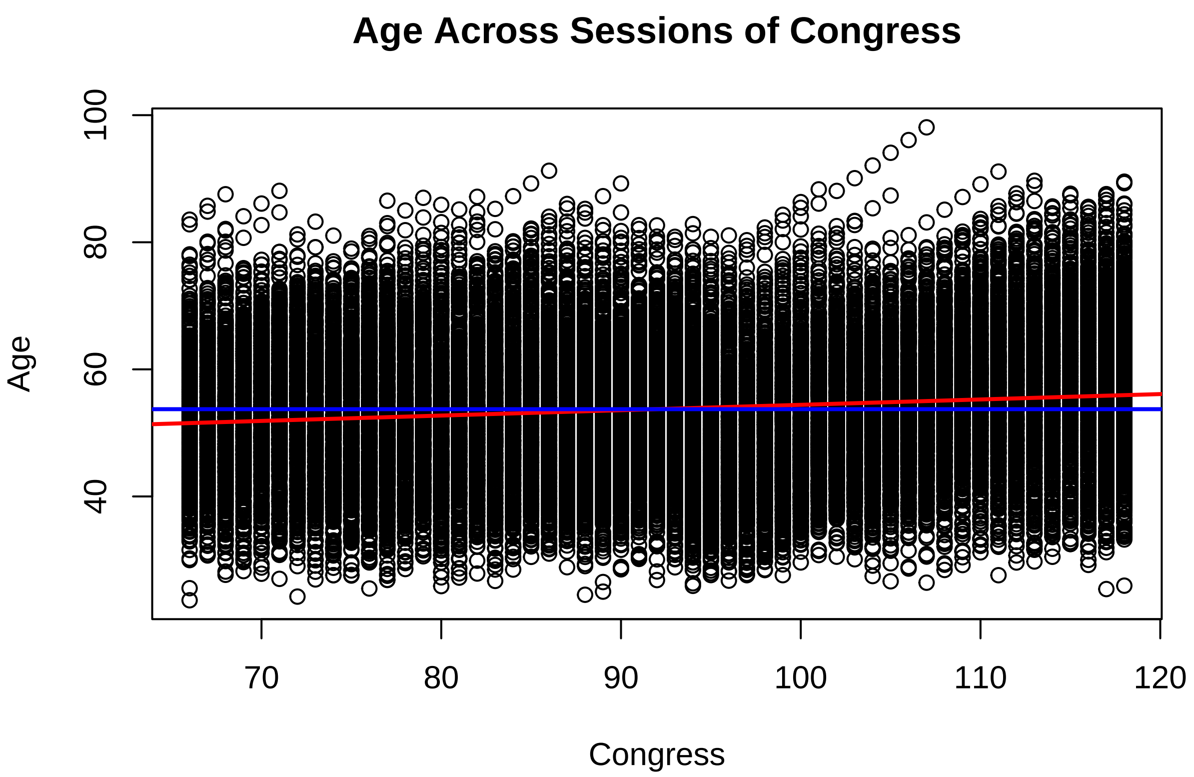
```
mean(oldc$age_years[oldc$party_code==200]) #average for Republicans
```

```
## [1] 53.77444
```

```
#find the mean for the house
```

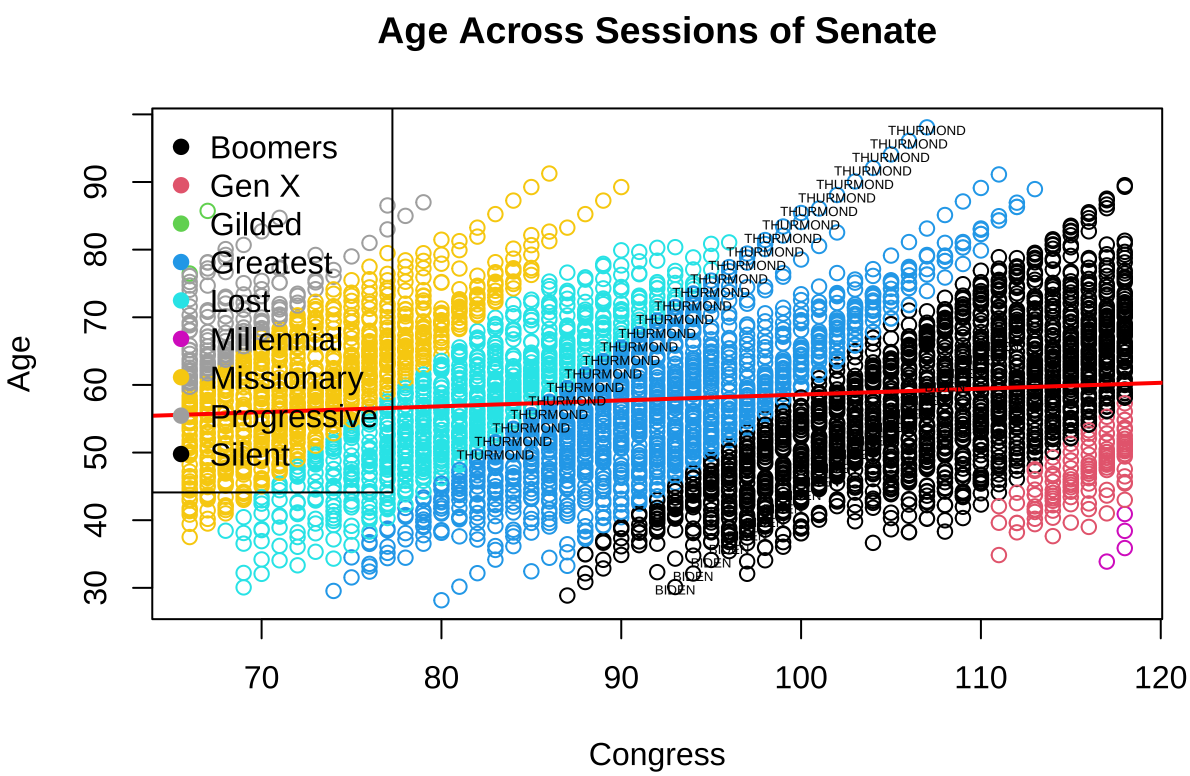
Let's make a simple scatterplot with congress # on the horizontal axis, and age on the vertical axis. A regression allows us to look at the relationship between variables linear regression model, and you're looking at the effect of the IV on the DV

```
plot(oldc$congress, oldc$age_years, ylab = "Age", xlab = "Congress", main = "Age Across Sessions of Cong
abline(lm(oldc$age_years ~ oldc$congress), col="red", lwd=2) # What type of relationship is this?
abline(h=mean(oldc$age_years), col="blue", lwd=2)
```



That is a bit overwhelming lets just look at the Senate - ie lets create a subset

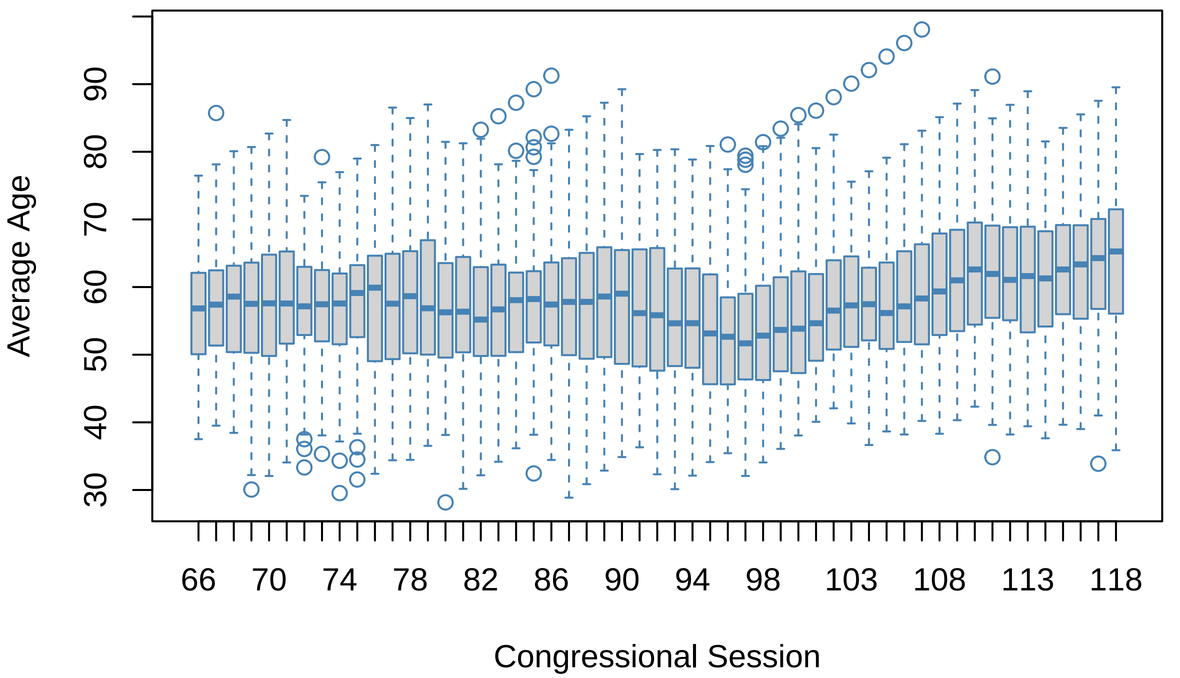
```
senate <- oldc[which(oldc$chamber=="Senate"),]

#New Graph
plot(senate$congress, senate$age_years, ylab = "Age", xlab = "Congress", main = "Age Across Sessions of
abline(lm(senate$age_years ~ senate$congress), col="red", lwd=2)
legend("topleft", legend = levels(factor(senate$generatio)), pch=19,col = factor(levels(factor(senate$ge

#I want to add labels for specific senators
#only want last time - take the characters prior to the comma
senate$lastname <- gsub(",.*$", "", senate$bioname)#You can ignore this line of code. I just googled ho
senate$stars <- ifelse(senate$lastname=="BIDEN"|senate$lastname=="THURMOND", senate$lastname, "") #crea
text(senate$congress, senate$age_years-.4, senate$stars, cex=.4)
```



Check: Can you create the same for the house

A boxplot for each session can show us how the distribution has changed over time

```
boxplot(senate$age_years~senate$congress, xlab = "Congressional Session", ylab="Average Age", border =
```



I've clearly had too much fun. Your turn to make something interesting.

Since we've created some new variables we may want to save our data:

```r
write.csv(senate, "senate.csv") #writes a csv to your working directory that you can open in excel
save(senate, file="senate.Rdata") #saves Rdata format to your working directory
```